Primer: RAG and Why It Sometimes Gets It Wrong

A five-minute read before Day 1. You do not need to remember anything here; the first 15 minutes of the session will show it on screen.

1. The problem

A language model only knows what it was trained on. It has never seen your campus records, your tickets or your fee notices. Ask it about them and it will either say it does not know, or it will produce something that sounds right.

2. RAG in one idea

Retrieval-Augmented Generation gives the model the relevant text at question time. The steps are:

1. Turn every record into a vector (a list of numbers that captures its meaning).
2. Turn the question into a vector too.
3. Find the records whose vectors are closest to the question.
4. Paste those records into the prompt and ask the model to answer from them.

The search in step 3 finds records that are about the same thing as the question. That is what vectors are good at.

3. Where it goes wrong

"About the same thing" is not the same as "connected to the same thing". Two records can use similar words and have nothing to do with each other. Put them side by side in a prompt, ask a question that assumes they are linked, and a model that wants to be helpful may agree.

The result is a confident answer that nobody ever recorded. You will see this happen on Day 1.

4. A second kind of structure

Some facts are not about meaning at all. They are about **who or what appears where**. A graph stores those links explicitly, so you can ask: is there a recorded path between these two things, or not?

You will see that the two tools answer different questions. Vector search finds what is relevant. A graph tells you what is connected. Think about what each one cannot do.

5. A model is not always wrong

In the lab you put the plain model's answer next to the grounded one. Do not expect the plain model to fail every time. Often it answers correctly; sometimes it invents a link that nobody recorded. Your job is to report what you actually see, and to explain each result in terms of the identifiers you chose to link on. An honest "the baseline got this one right" scores as well as a failure found.

6. What to bring

A laptop, a Google account, and the 10-minute setup check from the Student Setup Guide. Nothing else.

Questions to think about

- If a retrieval system returns the right records, can the answer still be wrong?
- How would you know whether two records are really about the same thing?
- What would you want a system to say when the answer is simply "no record of that"?